

Coffee Shop Sales Excel Dashboard Project

50 Business Questions for Week 4 Excel Analysis

Business Scenario: You are working as a Junior Data Analyst / Business Intelligence Analyst for a coffee shop business. The business has sales transaction data from different store locations. Your task is to use Microsoft Excel to analyse the data, answer business questions, build an interactive dashboard, and provide clear recommendations to management.

Main business problem: Management wants to understand sales performance across store locations, products, dates, weekdays, transaction hours and customer spending patterns. Your dashboard should help the business identify top-performing stores and products, busy periods, and opportunities to improve revenue, staffing, stock planning and promotions.

Student instruction: You do not need to answer every question with a separate chart. Use these questions to guide your Excel formulas, PivotTables, PivotCharts, slicers, timeline, dashboard design and final business recommendations.

Dataset columns to inspect

Column	Meaning
transaction_id	Unique transaction identifier
transaction_date	Date of the sale
transaction_time	Time of the sale
transaction_qty	Quantity bought
store_location	Coffee shop branch/location
unit_price	Price of one item
product_category	Main product group
product_type	Specific product type
product_detail	Detailed product name

Understanding the Dataset

No.	Business question
1	How many rows and columns are in the Coffee Shop Sales dataset?
2	What does each column represent in the dataset?
3	Which columns are numerical, and which columns are text or category fields?
4	Which columns contain date and time information?
5	What are the different store locations in the dataset?
6	What are the different product categories in the dataset?
7	What date range does the dataset cover?
8	Are there any missing values or duplicate transaction records that need to be checked before analysis?

Revenue and Transaction Analysis

No.	Business question
9	How much revenue did each transaction generate after creating a Revenue column?
10	What is the total revenue across all transactions?
11	What is the average revenue per transaction?

No.	Business question
12	Which transaction generated the highest revenue?
13	Which transaction generated the lowest revenue?
14	How many transactions are low-value, medium-value and high-value transactions?
15	What percentage of all transactions are high-value transactions?

Store Performance

No.	Business question
16	Which store location generated the highest total revenue?
17	Which store location generated the lowest total revenue?
18	Which store had the highest number of transactions?
19	Which store sold the highest total quantity of products?
20	Which store has the highest average transaction value?
21	Is the highest revenue store also the store with the highest number of transactions?
22	Which store may need further investigation or support from management?

Product Category and Product Performance

No.	Business question
23	Which product category generated the highest total revenue?
24	Which product category generated the lowest total revenue?
25	Which product category had the highest number of transactions?
26	Which product category sold the highest quantity of items?
27	Which product category has the highest average revenue per transaction?
28	What are the top 10 products by total revenue?
29	What are the top 10 products by quantity sold?
30	Which products should the business consider promoting more?

Date and Time Analysis

No.	Business question
31	Which month generated the highest total revenue?
32	Which month generated the lowest total revenue?
33	Is revenue increasing or decreasing month by month?
34	Which weekday generated the highest revenue?
35	Are weekday sales stronger than weekend sales?
36	What hour of the day has the highest number of transactions?
37	What hour of the day generates the highest revenue?
38	Does the business perform better in the morning, afternoon or evening?
39	Which hours may need more staff based on transaction volume?

Store and Product Combination Analysis

No.	Business question
40	Which store sells the most coffee products?
41	Which store sells the most tea products?
42	Which store sells the most bakery products?
43	Which product category performs best in each store location?
44	Which store-product category combination generates the highest revenue?
45	Does product performance differ by store location?

Dashboard, Visualisation and Recommendations

No.	Business question
46	Which KPI cards should appear at the top of the dashboard to help management understand performance quickly?
47	Which chart is best for comparing revenue by store location, and why?
48	Which chart is best for showing monthly revenue trend, and why?
49	What are the three most important insights from your analysis?
50	What three practical recommendations would you give to the coffee shop management?

Minimum dashboard deliverables

- Create a calculated Revenue column using transaction_qty multiplied by unit_price.
- Create useful date/time columns such as Month_Name, Weekday, Quarter, Transaction_Hour, Day_Type and Time_Band.
- Build PivotTables to summarise revenue, transactions, quantities and average transaction value.
- Create clear PivotCharts for store performance, product performance, monthly trend, hourly transactions and top products.
- Add slicers such as store_location, product_category and Day_Type, plus a timeline using transaction_date.
- Write at least three business insights and three recommendations for management.

Answer frame for each chart

For every chart you create, write a short explanation using this structure:

This chart answers the question:
I selected this chart because:
The key insight from this chart is:
This matters to the business because: